Problem 00000001876: a genuine finite definable group

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Result and exact scope. Both language versions define the relevant groups by dp-rank ≤ 1 in dependent theories and assert that all are solvable. There is no infinitude or connectedness premise. The actual finite group

$$G = S_5 = \mathrm{Perm}(\{0, 1, 2, 3, 4\})$$

is a counterexample. We build a real first-order group structure and a complete dependent theory in which this entire group and its operations are definable, prove the standard two-formula dp-minimality criterion in every model, and use actual nonsolvability of S_5.

The formalization uses the published pattern characterization of dp-minimality and dependence, not a general numerical dp-rank function. The example belongs to the finite cases allowed by the printed ≤ 1 definition. It is not a counterexample to an infinite-only or connected-group formulation, nor to a solvable-by-finite or nilpotent-by-finite classification. Those qualifications materially change the claim.

1. A genuine first-order group and its complete theory

The base language has an actual nullary identity symbol, unary inverse symbol and binary multiplication symbol, interpreted by the genuine operations of S_5. We expand it by one constant for each actual element of G. This is permitted by the source, which imposes no pure-language restriction. The domain is defined by $x = x$. Explicit formulas defining the unit, inverse graph and product graph satisfy

$$\mathrm{Unit}(x) \iff x = 1, \quad \mathrm{Inv}(x, y) \iff x^{-1} = y, \quad \mathrm{Mul}(x, y, z) \iff xy = z.$$

Lean proves these statements for actual `Formula.Realize`, not for unrelated predicates on a finite set.

Let L be this language and $T = \mathrm{Th}_L(G)$ its actual complete theory. Lean proves T is complete and that G is a model. A genuine finite-disjunction sentence of T says

$$\forall x \bigvee_{g \in G} x = c_g.$$

In every T-model M, it makes the actual map $g \mapsto c_g^M$ surjective. Thus *every* model of T, in arbitrary universe size, is finite. A named constant also ensures nonemptiness. The argument does not infer a theory-level property merely from inspecting one base model.

2. Dependence with all finite tuple arities

We use the standard independence-pattern definition stated in [1], printed p. 1. For an arbitrary actual formula $\varphi(x, y)$, with any finite lengths of x and y, an independence pattern would give tuples a_i and b_s, indexed by $i \in \mathbb{N}$ and $s \subseteq \mathbb{N}$, such that

$$M \models \varphi(a_i, b_s) \quad \iff \quad i \in s.$$

If $a_i = a_j$, choosing $s = \{i\}$ forces $i = j$. Therefore $i \mapsto a_i$ injects $\mathbb{N}$ into a finite tuple power of M, which is impossible. Lean quantifies over all finite arities and all formulas and proves absence of this actual pattern in each model. Hence T is dependent (NIP); there is no finiteness predicate renamed as dependence.

3. The full two-formula dp-minimality criterion

The same primary definition [1], p. 1, uses a singleton variable x, arbitrary finite parameter tuples and two formulas $\varphi(x, y)$ and $\psi(x, z)$. An ICT pattern would provide sequences a_i, b_j and witnesses $c_{i,j} \in M$ with, for all $i, j, i', j' \in \mathbb{N}$,

$$\varphi(c_{i,j}, a_{i'}) \iff i = i', \qquad \psi(c_{i,j}, b_{j'}) \iff j = j'.$$

Both selected and unselected indices occur explicitly in the formal definition. Fixing $j = 0$, if $c_{i,0} = c_{k,0}$, the first row, evaluated at parameter a_i, gives $i = k$. This injects $\mathbb{N}$ into the actual finite model M, again impossible. Thus there is no such ICT pattern in any model of T.

This proves the standard singleton-variable dp-minimality criterion appropriate to the source's dp-rank-≤ 1 definition. We do not restrict parameter arities or omit the negative requirements at unselected indices. Indeed the obstruction excludes even arbitrary sequences, so a version additionally requiring mutual indiscernibility is excluded too. No theory of a fictitious rank assigned to a finite group is introduced.

4. Actual nonsolvability and the resulting counterexample

The group remains the actual permutation group on five letters throughout. Lean invokes the Mathlib theorem `Equiv.Perm.not_isSolvable_fin_5`.

It proves nonsolvability using actual nonidentity permutations persisting in every derived subgroup; the conclusion is the genuine `Group.IsSolvable` predicate, not a Boolean label based on the group order.

The final theorem collects the complete theory, its actual group model, dependence and dp-minimality in all models, the four real definability statements, and nonsolvability of this same group. Consequently the source's necessary assertion "all are solvable" fails under its unrestricted definition, so no classification satisfying that assertion can cover all its stated cases. We do not attempt a replacement classification or claim that an infinite nonsolvable dp-minimal group has been constructed.

For context, the distinction from virtual conclusions is substantial: Wagner [2], Definition 1 and the main result, discusses the same pattern notion and proves a virtually nilpotent conclusion. A finite group is automatically virtually nilpotent via its trivial subgroup, which is compatible with this example. That general theorem is not imported as a Lean axiom and is not needed for the proof.

5. Verification and attribution

The project uses actual first-order syntax, `Formula.Realize`, `completeTheory`, genuine finite-model maps and actual group operations. All 39 explicit named declarations and the three constructors of the new `GroupSymbol` datatype are included in the 42-target axiom audit, including both proof-bearing interpretation instances. The package contains a new root build, warning-as-error audit, full fresh kernel replay and source-bound receipts, pinned dependencies, exact original bilingual source, and complete LaTeX/PDF. No `sorry`, `native_decide` or extra axiom is used.

The finite-group observation is elementary; the formal work closes its actual theory, formula-pattern and definable-group bridges. No earlier same-ID submission was found in the recorded all-state repository search. Research, proof and document production were led by a Codex worker with separate AI supervision of source semantics and PDF layout. Human verification and organizer acceptance are not asserted.

References

[1] I. Kaplan, E. Levi and P. Simon, *Some remarks on dp-minimal groups*, fixed author version arXiv:1605.07867v2, 5 May 2017, Introduction, printed p. 1: dp-minimality and NIP pattern definitions.

[2] F. O. Wagner, *Dp-minimal groups*, arXiv:2403.12111v1, 2024, Definition 1 and main virtually nilpotent theorem. Cited for context only.